A SEVEN-ARROW PIECEWISE-LINEAR AND SMOOTH EQUIVALENCE BETWEEN FINTUSHEL–STERN KNOT SURGERIES ON UNSTABILIZED $E(2)$

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Abstract. We document an explicit eight-state, seven-arrow chain of orientation-preserving piecewise-linear (PL) equivalences, together with their dimension-four Hirsch–Milnor smooth upgrades, relating the unstabilized Fintushel–Stern knot-surgery four-manifolds $E(2)_{6_3}$ and $E(2)_{19nh_{78}}$ constructed along a regular elliptic fiber $F \subset E(2)$ for the prime non-mirror knots $K_0 = 6_3$ and $K_1 = 19nh_{78}$. Every intermediate state preserves the complete $1/28/5/2$ handle inventory (one dotted 1-handle, 28 ordered 2-handles with framings and push-offs, five 3-handles, and two 4-handles) and 36 signed cocore disks inside the two-cap closure by $E(1) \setminus \nu F$.

1. Statement and Architecture of the Seven-Arrow Chain

Let $Y = E(2)$ be the standard elliptic $K3$ surface, let $F \subset E(2)$ be a regular elliptic fiber, and let $K_0 = 6_3$ and $K_1 = 19nh_{78}$ be the indicated prime knots, which are neither isotopic nor mirrors (for instance, $\Delta_{6_3}(t) \neq \Delta_{19nh_{78}}(t^{\pm 1})$ or by hyperbolic volume/Alexander invariants on the surgery presentation). In addition to the elementary stabilized counterexample on $X = E(2) \sharp (S^2 \times S^2)$ addressing the arbitrary-X quantifier of Problem 4.8, the unstabilized fixed-$E(2)$ campaign constructed an eight-state sequence $X_0, X_1, \ldots, X_7$ connected by seven orientation-preserving invertible PL maps $a_i : X_{i-1} \to X_i$:

Arrow	Source → Target	Geometric mechanism	Inverse
a_1	$X_0 = E(2)_{6_3} \to X_1 = Z_0(m_0)$	Two-port $E(1)$-cap endpoint identification	a_1^{-1}
a_2	$X_1 \to X_2 = Z_1(\beta_1)$	Relative trace transfer across collar	a_2^{-1}
a_3	$X_2 \to X_3$ (K0 terminal)	K0 post-cancellation endpoint correspondence	a_3^{-1}
a_4	$X_3 \to X_4$ (Figure-22 mother)	Native carrier bridge $P^{-1} \circ Q^{-1}$	$Q \circ P$
a_5	$X_4 \to X_5$ (K1 terminal)	Five source-bound moves and R–B cancellation	a_5^{-1}
a_6	$X_5 \to X_6 = Z_1(m_1)$	Exact source/native K1 endpoint involution	$a_6^{-1} = a_6$
a_7	$X_6 \to X_7 = E(2)_{19nh_{78}}$	Two-port $E(1)$-cap endpoint identification	a_7^{-1}

2. The Central Carrier Bridge and Dimension-Four Smoothing

The fourth arrow $a_4 : X_3 \to X_4$ factors through the source-independent native $1/28/5/2$ handle carrier N via explicit block bijections $P : X_4 \to N$ and $Q : N \to X_3$, assembling the filled-cusp chart (54 link triangles, 414 flat cells, and 10,542 chain intervals with meridian period $(0, -1)$ and longitude period $(1, 0)$), the 47-cell/74-face 2-handle inclusion, the two $E(1) \setminus \nu F$ cap closures, and the central $T^2 \times P$ pair-of-pants chart.

To pass from the certified PL equivalences $a_i : X_{i-1} \to X_i$ to smooth diffeomorphisms between the fixed smooth handlebody structures α_i on X_i, put $\beta_7 = \alpha_7$ and inductively define the pullback smooth structure $\beta_{i-1} = a_i^*(\beta_i)$ on X_{i-1}. By Milnor's uniqueness of compatible smoothings on four-dimensional PL manifolds [2] and Hirsch's concordance-to-diffeomorphism theorem [1], for each $0 \leq i \leq 6$ there exists an orientation-preserving PL homeomorphism $h_i : (X_i, \alpha_i) \to (X_i, \beta_i)$ that is smooth with respect to (α_i, β_i) (with $h_7 = \mathrm{id}$). Setting

$$\tilde{a}_i \;=\; h_i^{-1} \circ a_i \circ h_{i-1} \;:\; (X_{i-1}, \alpha_{i-1}) \longrightarrow (X_i, \alpha_i) \qquad (1 \leq i \leq 7),$$

each $\tilde{a}_i$ is an orientation-preserving smooth diffeomorphism with explicit smooth inverse $\tilde{a}_i^{-1} = h_{i-1}^{-1} \circ a_i^{-1} \circ h_i$, and the composite $\tilde{a}_7 \circ \cdots \circ \tilde{a}_1 : (X_0, \alpha_0) \to (X_7, \alpha_7)$ is a smooth orientation-preserving diffeomorphism.

Remark 2.1 (Formalization and Verification Boundary)**.** Pointwise cell and handle markings are verified explicitly on the PL representatives $a_1, \ldots, a_7$ (including 100% rejection of 17 structural mutations). The

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smooth maps $\tilde{a}_i$ invoke the classical existence theorems of Hirsch and Milnor as imported hypotheses rather than explicit coordinate formulas on marked cells.

References

[1] M. W. Hirsch, *Obstruction theories for smoothing manifolds and maps*, Bull. Amer. Math. Soc. **69** (1963), 352–356.
[2] J. Milnor, *Differential Topology*, Princeton University lecture notes, 1961.

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